

Jacob Edwards, PhD

Data Scientist · Analyst

[portfolio](#) · [github](#) · [linkedin](#) · [google scholar](#)

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Data scientist and analyst with a decade running controlled experiments, drawing inference from messy data, and turning quantitative results into clear visual stories. Created and taught Columbia Psychology's first applied data science course, taking two student capstones to peer-reviewed publication. Author of 6 peer-reviewed papers and NIH F31 awardee. PhD, Columbia.

Core Stack: Python · SQL · R · *scikit-learn* · *pandas* · Git

EXPERIENCE

Graduate Research Scientist

Columbia University · New York, NY · 2018 – Dec. 2025

- Led NIH-funded research program of **controlled experiments across 11 study populations** to determine causal drivers of vocal behavior and auditory perception. Results published in *Journal of Neuroscience* (2025) [↗](#) and *Scientific Reports* (2026) [↗](#).
- Trained and deployed a 3-classifier ensemble (SVM, Random Forest, KDE) on **411M+ labeled observations** of time-series sensor data, eliminating hundreds of hours of manual review across 300+ experiments and standardizing labeling across 4 teams.
- Engineered end-to-end ETL pipelines processing **3.2 TB of time-series sensor data** from 5,600+ recording channels across 20 subjects. Feature extraction, aggregation, validation, and visualization layered for downstream analyses.
- Shipped self-service interactive visualization tools (3D dashboards, time-aligned audio + visual), enabling non-technical collaborators to explore complex time-series data as audiovisual “movies”; **adopted by 3 additional teams as standard tooling**.
- Built predictive sequence models (NLP-style; custom fuzzy-match evaluation metric and bootstrapped confidence intervals) on **30k+ tokenized acoustic events from 180 subjects**.
- Unified database and annotation infrastructure via custom GUI, **cutting work hours by 80%** on similar projects across 3 teams.
- Trained incoming lab members (postdocs, graduate and undergraduate students) on the lab's analysis pipelines and experimental methods. Mentored 5 undergraduates; three went on to **Yale Medicine**, the **NY Genome Center**, and a **Fulbright**.
- Raised **\$150k in competitive research funding**, including an **NIH (F31) predoctoral grant** [↗](#).

Statistics & Data Science Course Designer/Instructor

Columbia University · New York, NY · 2020 - 2024

- Designed and taught Columbia Psychology's first [applied data science course](#) [↗](#), covering R, data wrangling, visualization, machine learning, NLP, network analysis, and **algorithmic fairness**. 20 hours of original lectures and 7 hands-on assignments.
- Built and taught civic-data analyses with **real public datasets**: NYC OpenData, ProPublica COMPAS, NIJ recidivism, Stanford Open Policing, Snow's 1854 cholera map.
- Mentored **two student capstones to peer-reviewed publication**: a geospatial analysis of NYPD use-of-force and domestic violence in *Violence Against Women*; a risk analysis of cybersecurity breaches and AI in *Columbia Undergraduate Law Review*.
- Three students used the **course as a launchpad** to graduate programs at LSE, NYU (Urban Data Science), and Columbia's Center for Justice. Highly rated by students in evaluations.

Independent Data Science Work

New York, NY · Jan. 2026 - present

- Built a public-facing [portfolio](#) [↗](#) of applied ML, statistical modeling, and data visualization on neural, acoustic, and civic datasets (incl. NYC OpenData), with production write-ups and code.

SELECTED PROJECTS

Do 311 calls actually represent New Yorkers' problems?

NYC Reporting Gap Index · [write-up](#) [↗](#) · [GitHub](#) [↗](#) · [Tableau](#) [↗](#)

Python · SQL · R · Tableau · civic data · regression modeling

- Generalizable analytic pipeline linking 6 NYC open datasets to test whether 311 reports diverge from DOT operational records across 59 community districts. Negative binomial regression with bootstrap inference. Citizen reports and city repairs were highly concordant ($r=0.88$), with no district showing a statistically significant gap.

SpikeJungle

Production ML Classifier on Time-Series Data · [write-up](#) [↗](#) · [GitHub](#) [↗](#)

MATLAB · ensemble methods · feature engineering · time-series

- 3-model ensemble (SVM, Random Forest, KDE) automating noise removal from neural voltage recordings; trained on 411M+ labeled observations across 2 published studies. Validated on 32 unseen experiments with 74-92% inter-classifier agreement. Preserved biological signal post-cleaning. Injected into standard ETL pipeline with a single-line toggle for ease of use.

SKILLS

Languages & tools	Python (pandas, NumPy, scikit-learn, matplotlib), R (tidyverse, ggplot2), SQL, Tableau, MATLAB, version control (Git, GitHub), notebooks (Jupyter, RStudio), regex
ML & statistics	Regression & classification, time-series analysis, geospatial analysis, NLP & sequence modeling, dimensionality reduction, clustering, feature engineering, hyperparameter tuning, cross-validation
Experimentation	Experimental design, causal inference, hypothesis testing, A/B test methodology, bootstrapping, effect size estimation, simulation, metric design
Visualization & design	Data visualization with intention, precision illustrations, interactive dashboards and GUIs, multi-panel composition, color theory, typography, Tufte principles (chartjunk, data-ink ratio, small multiples)
AI workflows	Claude (Claude Code, projects, skills), prompt engineering for writing, research & code review

PUBLICATIONS & PRESENTATIONS

1. **Edwards JA**, Woolley SMN. (2026). [Models of species' syntax rules predict vocal sequences](#). *Scientific Reports* **16**: 14795
2. **Edwards JA**, Rivera M, Woolley SMN. (2025). [Temporal behavior is driven by species](#). *Journal of Neuroscience* **45**: e0576242025
3. Rivera M, **Edwards JA**, ..., Woolley SMN. (2023). [Machine learning links vocal acoustics to ancestry](#). *Scientific Reports* **13**: 7076
4. **Edwards JA**, Risch MN, Hoke KL. (2021). [PNN dynamics over metamorphosis](#). *Journal of Comparative Neurology* **529**: 1768-1778
5. So NLT, **Edwards JA**, Woolley SMN. (2020). [Auditory selectivity in brain and behavior](#). *Journal of Neuroscience* **40**: 1015-1027
6. Williams JH, **Edwards JA**, Ramsey AJ. (2016). [Economy and evolution of pollen growth](#). *American Journal of Botany* **103**: 471-483
 - Data visualization selected by editorial staff for [journal cover image](#) ↗

Selected Presentations: Society for Neuroscience (2019, 2022), Taste of Science NYC (2021), Frontiers of Neuroscience, Barnard College (2021), BioBus Harlem (2018, 2019)

EDUCATION

PhD, Psychology (Auditory Neuroscience)
MSc, Biology
BSc, Biology

Columbia University · 2025
Colorado State University · 2018
University of Tennessee, Knoxville · 2013